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Labeling machine

Abstract

A labeling machine includes a conveyor of containers to be labeled along an advancement path and at least one labeling station. The labeling station includes an unreeling assembly for a label ribbon, and a cutting drum. At least one cutting device is provided and has a respective suction chamber, which accommodates internally a blade. The suction chamber is connectable with an air suction device, in order to attract the label ribbon toward the inside of the suction chamber and against the blade, in order to cut the label ribbon. The cutting device includes a respective blade supporting element which is coupled to the cutting drum and supports the blade. This blade supporting element is elastically deformable in order to allow the coupling/uncoupling of the blade to/from the blade supporting element.

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Background/Summary

(1) The present invention relates to a labeling machine.
(2) Labeling machines are known which make it possible to apply, on containers, labels that are obtained by cutting a continuous label ribbon which is taken from a reel on which a plurality of labels are printed in succession.
(3) The label ribbon can be provided with strips of glue pre-applied to the face thereof that is intended to come into contact with the containers, or it can be without glue and, in this case, the machine is provided with a gluing unit, which is designed to apply the glue on the labels.
(4) Generally, labeling machines comprise a conveyor, typically constituted by a rotating carousel, that transports the containers to be labeled along an advancement path, and at least one labeling station, which is designed to apply the labels to the containers in transit on the conveyor and which is usually provided with an unreeling assembly, which is designed to take the label ribbon from its reel and to feed it to a rotating cutting drum, which is designed to receive the label ribbon so that it adheres to its own lateral surface.
(5) In some labeling machines, the cutting drum is provided with blades that are integral therewith, and which, by interacting with a stationary blade placed laterally to the cutting drum, make it possible to perform the cut of the label ribbon at the separation region between two labels consecutive.
(6) The labels separated after cutting are then transferred from the cutting drum to a transfer drum which places them so that they adhere on the respective container arriving on the carousel.
(7) The cutting drum described above has the drawback that it requires a precise adjustment of the position of the blades that are integral with the cutting drum relative to the stationary blade, in order to ensure the correct execution of the cutting of the labels.
(8) In other labeling machines, the cutting drum is provided with blades that are made to move by respective working cylinders, in order to allow the cutting of the label ribbon to be carried out without requiring the presence of the stationary blade.
(9) In this case, the cutting drum also transfers the label, separated after cutting, directly to the corresponding container and is, therefore, more correctly termed “cutting and transfer drum”.
(10) These latter machines, although they have fewer problems with adjusting the blades, do however feature a greater complexity of construction over the machines described previously.
(11) In order to overcome these and other drawbacks in the known art, a labeling machine has been proposed, described in Italian patent no. 1423941, in the name of this same applicant, in which the cutting drum is provided, along its perimeter, with a plurality of cutting devices, which are angularly mutually spaced apart and are each constituted by a respective blade which is integrally fixed to the cutting drum and accommodated in a respective suction chamber which is defined inside the cutting drum and is open at the lateral surface of the cutting drum.
(12) The suction chamber of each cutting device can be connected to air suction means, so as to enable the attraction, toward the inside of the suction chamber, of the portion of the label ribbon that in each instance is directed toward the suction chamber, so as to be able to push it against the blade, thus providing the cutting thereof.
(13) The machine just described, although it does not necessitate any operation to adjust the position of the blades and it is simple to provide in terms of construction, entails a certain inconvenience in operations to replace the blades, once they are worn.
(14) In particular, extracting the blades from the suction chamber is quite a complex task, and consequently the time required to carry out this operation is rather long.
(15) The aim of the present invention is to provide a labeling machine that is capable of improving the known art in one or more of the above mentioned aspects.
(16) Within this aim, an object of the invention is to provide a labeling machine that does not require exact adjustments of the blades and which at the same time allows their simple and convenient replacement.

(17) Another object of the invention is to provide a labeling machine that in terms of construction is simple to provide.
(18) Another object of the present invention is to provide a labeling machine that can offer the widest guarantees of reliability and safety in its operation.
(19) A further object of the present invention is to overcome the drawbacks of the background art in a manner that is alternative to any existing solutions.
(20) Another object of the invention is to provide a labeling machine that can be produced at low cost, so as to be competitive from a purely economic viewpoint as well.
(21) This aim and these and other objects which will become better apparent hereinafter are achieved by a labeling machine according to claim **1**, optionally provided with one or more of the characteristics of the dependent claims.

Description

(1) Further characteristics and advantages of the invention will become better apparent from the description of a preferred, but not exclusive, embodiment of the labeling machine according to the invention, which is illustrated by way of non-limiting example in the accompanying drawings wherein:
(2) FIG. **1** is a perspective view of the labeling machine according to the invention;
(3) FIG. **2** is a perspective view of a cutting drum of the labeling machine according to the invention;
(4) FIG. **3** is an exploded perspective view of a cutting device of the machine according to the invention uncoupled from the corresponding cutting drum and with the blade supporting element in the deformed condition;
(5) FIG. **4** is a front elevation view of the cutting device uncoupled from the cutting drum with the blade supporting element in the non-deformed condition;
(6) FIG. **5** is a front elevation view of the cutting device uncoupled from the cutting drum with the blade supporting element in the deformed condition.
(7) With reference to the figures, the labeling machine according to the invention, generally designated by the reference numeral **1**, comprises a conveyor **2** that moves containers **3** to be labeled along an advancement path.
(8) For example, the conveyor **2** is constituted by a carousel **4**, which rotates about its own axis and which has, on its peripheral region, a plurality of pans **5** for supporting the individual containers **3**, with which the containers are rotated about their axes.
(9) At least one labeling station **6** is arranged along the advancement path of the containers **3**, and comprises an unreeling assembly **7**, which is constituted, for example, by a pair of mutually opposite entrainment rollers **7***a*, **7***b* which is designed to take a label ribbon **8**, on which a plurality of labels **8***a* are printed continuously, from a reel, not shown, making the label ribbon **8** pass, conveniently, through a tension adjustment device **9**, as is known per se.
(10) The labeling station **6** is further provided with a cutting drum **10**, which can move by rotation about its axis and is designed to receive, so that it adheres to the lateral surface **10***a* thereof, the label ribbon **8**.
(11) To this end, on the lateral surface **10***a* of the cutting drum **10** there are holes **11** connected to air suction means, which are constituted, for example, by a vacuum pump **100**, shown schematically, and which make it possible to hold the label ribbon **8** against the lateral surface **10***a* of the cutting drum **10**.
(12) In addition, at least one cutting device **12** is arranged along the perimeter of the cutting drum **10** and is provided with a respective suction chamber **13**, which is open at the lateral surface **10***a* of the cutting drum **10** and internally accommodates a blade **14**, which is designed to cut the label

ribbon **8** and which lies, conveniently, on a radial plane with respect to the axis of the cutting drum **10**.
(13) More specifically, the suction chamber **13** can be connected to air suction means, for example constituted by the same vacuum pump **100** to which the holes **11** are connected, in such a way as to be able to attract the label ribbon **8** toward the inside of the suction chamber **13**, so as to push it against the blade **14** and, as a consequence, cut the label ribbon **8**.
(14) According to the invention, the (or each) cutting device **12** comprises a respective blade supporting element **15**, which is coupled, preferably removably, to the cutting drum **10** and supports the blade **14**.
(15) Also according to the invention, this blade supporting element **15** is elastically deformable in order to allow the coupling/uncoupling of the blade **14** to/from the blade supporting element **15**.
(16) As in the example shown, along the peripheral region of the cutting drum **10**, there can optionally also be multiple cutting devices **12**, which are arranged at a circumferential distance from each other that is at least equal to the length of the labels **8***a*.
(17) In this case, at least one and, more preferably, each one of the cutting devices **12** is provided with a corresponding blade supporting element **15**.
(18) In greater detail, the blade supporting element **15** of each cutting device **12** is, advantageously, constituted by a box-like body **15***a*, which internally defines the suction chamber **13** of the corresponding cutting device **12** and to which the corresponding blade **14** is coupled.
(19) This box-like body **15***a* is made of an elastically deformable material, such as, for example, rubber or the like, so that as a consequence of its deformation, it is possible to uncouple the blade **14** therefrom and extract it from the suction chamber **13**.
(20) In particular, in order to allow the coupling of the blade **14** to the box-like body **15***a*, on the box-like body **15***a* there is at least one locking pin **16** which removably engages a corresponding retention opening **17**, which is defined in the blade **14** and, preferably, is elongated in shape, so as to facilitate the engagement thereof by the locking pin **16**.
(21) This locking pin **16** is, in practice, disengageable from the retention opening **17**, so as to free the blade **14** from the coupling with the box-like body **15***a*, following an elastic deformation imposed on the box-like body **15***a* that is such as to produce the extraction of the locking pin **16** from the corresponding retention opening **17**.
(22) Preferably, on the box-like body **15***a* there are multiple locking pins **16**, arranged mutually spaced apart along a direction that is substantially parallel to the axis of the cutting drum **10** and, therefore, to the longitudinal extension of the blade **14**.
(23) The locking pins **16** can be made in a single piece with the box-like body **15***a* or they can be constituted by inserts made of a different material from the material with which the box-like body **15***a* is provided, and which are partially embedded in the box-like body.
(24) In more detail, the box-like body **15***a* is provided, inside the suction chamber **13**, with a pair of mutually facing walls **18***a*, **18***b*, between which the blade **14** is inserted, at least partially.
(25) In particular, the walls **18***a*, **18***b* are arrangeable further apart from each other as a consequence of the elastic deformation of the box-like body **15**.
(26) More specifically, the locking pins **16** are defined on at least one of the walls **18***a*, **18***b*, so that by virtue of the spacing apart of the walls **18***a*, **18***b*, which is achieved as a result of the elastic deformation of the box-like body **15***a*, it is possible to obtain the disengagement of the locking pins **16** from the retention openings **17** of the blade **14**, so as to obtain the decoupling of the blade **14** from the box-like body **15***a*.
(27) Advantageously, on the walls **18***a*, **18***b* there are abutment protrusions **19**, which are designed to engage by resting against the opposite faces of the blade **14**, in order to ensure the correct positioning of the blade **14** with respect to the box-like body **15***a*, in the active condition.
(28) It must be noted that the box-like body **15** is coupled to the cutting drum **10** and, conveniently, it can be extracted from or inserted into a respective accommodation seat **20**, which is defined

peripherally in the cutting drum **10**, by way of its sliding along a direction that is substantially parallel to the axis of the cutting drum.

(29) It should be noted that, in order to allow the ingress of the label ribbon **8** inside the suction chamber **13**, the box-like body **15***a* has, on its face intended to be directed outward, a suction opening **13***a*, which communicates with the suction chamber **13** and which lies, when the box-like body **15***a* is inserted in its accommodation seat **20**, substantially at the lateral surface **10***a* of the cutting drum **10**.

(30) Conveniently, on the other side with respect to the suction opening **13***a*, the box-like body **15***a* is provided with a coupling portion **22**, which is substantially dovetail-shaped and which removably engages a correspondingly-shaped portion **23** of the accommodation seat **20**, in order to guide the sliding movement of the box-like body **15***a* in the direction parallel to the axis of the cutting drum **10** that allows its insertion into and extraction from the accommodation seat **20** and in order to block, at the same time, the possibility of the box-like body **15***a* to move in the radial direction, with respect to the axis of the cutting drum **10**, when the box-like body **15***a* is inserted in the accommodation seat **20**.

(31) It must be noted that, as in the example shown, it is possible for each cutting device **12** of the cutting drum **10** to be flanked by a respective skid **24**, which is arranged in position immediately before the accommodation seat **20** for the box-like body **15***a* of the corresponding cutting device **12**, in the direction of rotation of the cutting drum **10**, and which protrudes slightly from the lateral surface **10***a* of the cutting drum **10**, in order to allow, in cooperation with a glue roller **25** positioned laterally to the cutting drum **10**, the application of strips of glue on the trailing edge and on the leading edge of two consecutive labels **8***a* on the label ribbon **8**, before the label ribbon itself is cut by the cutting device **12** adjacent thereto.

(32) The operation of the labeling machine, according to the invention, is the following.

(33) In an initial situation, a skid **24**, on which the trailing edge and the leading edge of two consecutive labels **8***a* of the label ribbon **8** are resting, passes in front of the glue roller **25**, so that the glue roller **25** can apply strips of glue on the trailing edge and on the leading edge of these labels **8***a*, and the unwinding assembly **7** feeds the label ribbon **8** to the cutting drum **10** at a speed that corresponds substantially to the peripheral speed of the cutting drum **10**.

(34) Subsequently, the unwinding assembly **7** feeds the label ribbon **8** to the cutting drum **10** at a speed slower than the peripheral speed of the cutting drum **10**, so as to obtain the positioning, at the separation region between the two labels **8***a* on which the strips of glue have been applied, of the cutting device **12** immediately behind, in the direction of rotation of the cutting drum **10**, the skid **24** that in the previous step was at the glue roller **25**.

(35) Once this cutting device **12** has reached the separation region between the two labels **8***a*, the label ribbon **8** is fed again by the unwinding assembly **7** at a speed that substantially corresponds to the peripheral speed of the cutting drum **10**, and the suction chamber **13** of this cutting device **12** is connected to the air suction means, so as to attract the label ribbon **8** toward the inside of the suction chamber **13** and push it against the corresponding blade **14**, so obtaining the cutting of the label ribbon **8**, with separation therefrom of one of the labels **8***a* from the rest of the label ribbon **8**.

(36) At this point, the unwinding assembly **7** feeds the label ribbon **8** to the cutting drum **10** at a speed slower than the peripheral speed of the cutting drum **10**, in so doing making the label **8***a* thus cut from the rest of the label ribbon **8** from which it has been separated move away, and until it is possible to position a skid **24** at the trailing edge and the leading edge of the next two consecutive labels **8***a* of the label ribbon **8**, after which the unwinding assembly **7** will again feed the label ribbon **8** to the cutting drum **10** at a speed that substantially corresponds to the peripheral speed of the cutting drum **10**.

(37) Subsequently, by proceeding in its rotation, the cutting drum **10** will transfer the label **8***a* thus cut to the respective container **3** transported by the conveyor **2**.

(38) If it becomes necessary to replace one of the blades **14** of the cutting devices **12** of the cutting

drum **10**, with the machine stopped the operator uncouples the corresponding box-like body **15***a* from the cutting drum **10**, by making it slide along a direction that is substantially parallel to the axis of the cutting drum **10**, so as to obtain the extraction thereof from the corresponding accommodation seat **20**.
(39) Once the box-like body **15***a* has been extracted from its accommodation seat **20**, the operator uncouples the blade **14** from the box-like body **15***a*, first making the box-like body **15***a* undergo an elastic deformation, with mutual spacing apart of the walls **18***a*, **18***b*, so as to obtain the disengagement of the locking pins **16** from the retention openings **17**, and then extracting the blade **14** from the suction chamber **13**, through the suction opening **13***a*.
(40) At this point, while keeping the box-like body **15***a* in the deformed condition, a new blade **14** can be inserted in the suction chamber **13** of the box-like body **15***a* and arranged between the walls **18***a*, **18***b*, and then allowing the box-like body **15***a* to elastically return to the non-deformed condition so that the locking pins **16** can insert into the relative retention openings **17**, thus obtaining the coupling of the blade **14** to the box-like body **15***a*.
(41) At this point the box-like body **15***a* can again be coupled to the cutting drum **10** via the insertion thereof into the corresponding accommodation seat **20**.
(42) In practice it has been found that the invention fully achieves the intended aim and objects, in that, by virtue of the capacity for elastic deformation of the blade supporting elements of the cutting drum, it makes it possible to facilitate the operations to replace the blades of the cutting drum and make them extremely rapid.
(43) It should further be noted that the capacity for elastic deformation of the blade supporting element and, more specifically, of the corresponding box-like body enables an easy cleaning of the suction chamber, in addition to ensuring a better seal thereof with the label ribbon.
(44) Another advantage that it is possible to achieve by virtue of the invention consists of the fact that the coupling of the box-like body made of elastically deformable material with the corresponding accommodation seat defined in the cutting drum entails fewer problems from the point of view of pneumatic leaks, compared to the known art, in that the seal is adequately ensured by the deformability of the box-like body, without the need to use any additional gasket.
(45) The invention thus conceived is susceptible of numerous modifications and variations, all of which are within the scope of the appended claims. Moreover, all the details may be substituted by other, technically equivalent elements.
(46) In practice the materials employed, provided they are compatible with the specific use, and the contingent dimensions and shapes, may be any according to requirements and to the state of the art.
(47) The disclosures in Italian Patent Application No. 102021000004820 from which this application claims priority are incorporated herein by reference.

Claims

1. A labeling machine, comprising: a conveyor of containers to be labeled along an advancement path and at least one labeling station, which is arranged along said advancement path, said labeling station comprising an unreeling assembly for a label ribbon, on which a plurality of labels are printed continuously, and a cutting drum, which can move by rotation about an axis thereof and is designed to receive, so that it adheres to the lateral surface thereof, said label ribbon, at least one cutting device being arranged along a perimeter of said cutting drum and having a respective suction chamber, which is open at a lateral surface of said cutting drum and accommodates internally a blade, said suction chamber being connectable to air suction means, in order to attract said label ribbon toward an inside of said suction chamber and against said blade, in order to cut said label ribbon, wherein said at least one cutting device comprises a respective blade supporting element which is coupled to said cutting drum and supports said blade, said blade supporting element being elastically deformable in order to allow coupling/uncoupling of said blade to/from

said blade supporting element; wherein said blade supporting element comprises a box-like body which defines said suction chamber internally, said box-like body being made of an elastically deformable material.

2. The machine according to claim 1, wherein said blade supporting element is coupled to said cutting drum removably.

3. The machine according to claim 1, wherein at least one locking pin is defined on said box-like body and removably engages a corresponding retention opening defined in said blade, said at least one locking pin being disengageable from said retention opening following an elastic deformation imposed on said box-like body.

4. The machine according to claim 3, wherein said box-like body has, inside said suction chamber, a pair of mutually facing walls, said blade being inserted, at least partially, between said walls, said pair of mutually facing walls being arrangeable further apart from each other as a consequence of said elastic deformation.

5. The machine according to claim 4, wherein said at least one locking pin is defined on at least one of said pair of mutually facing walls.

6. The machine according to claim 4, further comprising, on said pair of mutually facing walls, abutment protrusions which engage by resting against opposite faces of said blade.

7. The machine according claim 1, wherein said box-like body is configured to be extracted from or inserted into a respective accommodation seat, which is defined peripherally in said cutting drum, by way of sliding along a direction that is substantially parallel to the axis of said cutting drum.

8. The machine according to claim 7, wherein said box-like body has, on a face thereof intended to be directed outward, a suction opening which communicates with said suction chamber and lies, with said box-like body inserted in said accommodation seat, substantially at the lateral surface of said cutting drum.

9. The machine according to claim 8, wherein said box-like body has, on an opposite side with respect to said suction opening, a coupling portion which is substantially dovetail-shaped and removably engages a correspondingly-shaped region of said accommodation seat.